

Question number	Answer	Additional guidance	
2(a)	Choose an item. • C is the correct answer A is not the correct answer as CT does not show changes in brain function as they happen. B is not the correct answer as ECG does not show brain function. D is not the correct answer as PCR does not show brain function.		(1)

number			
2(b)	An answer that includes three of the following points: • advantage of MRI (1) • • advantage of PET (1) • disadvantage of MRI (1) • disadvantage of PET (1)	if factors refer to both and one wrong and one right ignore wrong one e.g MRI gives detailed image, 3D image, no(ionizing) radiation, good soft tissue contrast, high resolution, can see diseased tissue uses magnetic field so safe for pregnant women ignore safe for all individuals e.g. PET detects biochemical changes in brain, can observe cancer growth, used to diagnose dementia, shows which parts of brain are {active/ metabolizing} PET scan 3D e.g. difficult for people who dislike enclosed spaces, long time, some need sedation, expensive, doesn't show activity of brain, limiting accessibility in some countries? noisy/cant be used on patient with pacemaker or metal implant e.g. exposure to (gamma / beta) radiation, uses radioactive tracer, lower resolution, expensive, 2D image	(3)

Question number	Answer	Additional guidance	Mark
2(c)	Choose an item. • stated function cerebellum (1) • stated function of cerebral hemisphere (1) • stated function of hypothalamus (1)	(coordination of voluntary) movements, posture/ balance and speech ignore co-ordination unqualified (control of) voluntary behaviour / site of intelligence/ memory / learning / thinking/ emotions / imagination / voluntary response/ motor skills interpreting language, spatial and visual perception, control of voluntary movements homeostasis/ correctly named hormone production/thermoregulation/ controls pituitary gland accept osmoregulation	(3)

